

Identification of organic chemical indicators for tracking pollution sources in groundwater by machine learning from GC-HRMS-based suspect and non-target screening data

Okon Dominic Ekpe^a, Gyojin Choo^b, Jin-Kyu Kang^c, Seong-Taek Yun^d, Jeong-Eun Oh^{a,c,*}

^a Department of Civil and Environmental Engineering, Pusan National University, Busan 46241, South Korea

^b School of Natural Resources and Environmental Science, Kangwon National University, Chuncheon 24341, South Korea

^c Institute for Environment and Energy, Pusan National University, Busan 46241, South Korea

^d Department of Earth and Environmental Sciences, Korea University, Seoul 02841, South Korea

ARTICLE INFO

Keywords:

GC-HRMS
Suspect/non-target screening
Chemical indicators
Groundwater
Environmental forensics
Machine learning

ABSTRACT

In this study, the strong analytical power of gas chromatography coupled to a high resolution mass spectrometry (GC-HRMS) in suspect and non-target screening (SNTS) of organic micropollutants was combined with machine learning tools for proposing a novel and robust systematic environmental forensics workflow, focusing on groundwater contamination. Groundwater samples were collected from four different regions with diverse contamination histories (namely oil [OC], agricultural [AGR], industrial [IND], and landfill [LF]), and a total of 252 organic micropollutants were identified, including pharmaceuticals, personal care products, pesticides, polycyclic aromatic hydrocarbons, plasticizers, phenols, organophosphate flame retardants, transformation products, and others, with detection frequencies ranging from 3 % to 100 %. Amongst the SNTS identified compounds, a total of 51 chemical indicators (i.e., OC: 13, LF: 12, AGR: 19, IND: 7) which included level 1 and 2 SNTS identified chemicals were pinpointed across all sampling regions by integrating a bootstrapped feature selection method involving the *bootfs* algorithm and a partial least squares discriminant analysis (PLS-DA) model to determine potential prevalent contamination sources. The proposed workflow showed good predictive ability (Q^2) of 0.897, and the suggested contamination sources were gasoline, diesel, and/or other light petroleum products for the OC region, anthropogenic activities for the LF region, agricultural and human activities for the AGR region, and industrial/human activities for the IND region. These results suggest that the proposed workflow can select a subset of the most diagnostic features in the chemical space that can best distinguish a specific contamination source class.

1. Introduction

Groundwater is an important source of drinking water globally, representing the largest part of the world's available freshwater resources at 97 % (Guppy et al., 2018). Unfortunately, it is prone to environmental contamination by a wide range of organic micropollutants via diverse sources, including agricultural and industrial practices, runoff of animal waste, landfill leachate, improper disposal of oil-based products and household/personal care products, as well as wet deposition followed by seepage (Fries and Püttmann, 2004; Kayastha et al., 2022; Kurwadkar, 2017; Sibley et al., 2002). Once released into the environment, the organic contaminants can travel through the soil and reach groundwater, where they can persist for long periods of time.

Notably, the chemical analysis of groundwater can provide important information about the occurrence and distribution of organic compounds, which is a first step towards contaminants source tracking in the environment. However, conventional analytical methods have a limitation in identifying a limited number of target pollutants, potentially missing important chemical indicators not pre-selected for analysis. Therefore, there is a need for comprehensive and sensitive analytical approaches that can screen for both target (or suspect) and non-target pollutants, including both parent compounds and transformation products to obtain a holistic understanding of the contamination characteristics of samples (Gago-Ferrero et al., 2016; Mazur et al., 2021; Wang et al., 2020).

In light of the above, gas chromatography coupled to a high-

* Corresponding author at: Department of Civil and Environmental Engineering, Pusan National University, Busan 46241, South Korea.

E-mail address: jeoh@pusan.ac.kr (J.-E. Oh).

<https://doi.org/10.1016/j.watres.2024.121130>

Received 28 October 2023; Received in revised form 5 January 2024; Accepted 10 January 2024

Available online 13 January 2024

0043-1354/© 2024 Elsevier Ltd. All rights reserved.

resolution mass spectrometry (GC-HRMS) has emerged as a powerful tool for the identification of a wide range of known and unknown organic pollutants in sample extracts through suspect and non-target screening (SNTS). With GC-HRMS-based SNTS, large-scale compounds can be identified and characterized through the use of available GC accurate mass spectral libraries, which could potentially reduce the time and costs involved in acquiring reference standard materials for compounds that might not be detected in the analyzed samples (Ekpe et al., 2022; Moschet et al., 2017). Nevertheless, the large amounts of data generated from GC-HRMS-based suspect and non-target analysis require advanced computational methods to extract meaningful information. Hence, a combination of advanced GC-HRMS analysis and machine learning (ML) techniques can help to uncover the patterns of chemical constituents in environmental samples, which could lead to the identification of important chemical indicators that are indicative of specific contamination sources and pathways (Kalogiouri et al., 2020; Rivera-Pérez et al., 2022; Zhong et al., 2021).

The common goal of chemical forensic analysis is the selection of a panel of diagnostic chemical features for contamination source identification. While some previous studies have made efforts to identify and suggest the organic chemical indicators for tracing sources based on the chromatographic profile and compounds diagnostic ratios of the analyzed samples (Thomas and Delfino, 1991; Wang et al., 1999), other studies have attempted to utilize multivariate statistical tools, such as clustering analysis, co-occurrence analysis (e.g., venn diagrams), and principal component analysis for a similar purpose (Peter et al., 2018; Masiá et al., 2014). However, a common pitfall of these approaches is in their inability to pinpoint the most diagnostic chemical features that are important for source prediction, especially for chromatographic signals that are masked at low peak areas or detection frequencies. Recently, Kibbey et al. (2021) utilized multiple supervised ML models to address this limitation but their research only focused on binary classification problem and their approach of manually iterating through all possible combinations of chemical features is impractical for large datasets, such as SNTS data. To the best of our knowledge, studies combining HRMS-based SNTS with ML methods for identifying diagnostic chemical fingerprints have rarely been performed for the long-term field-scale tracking of pollutants contamination pathways in groundwater with diverse contamination histories, including both oil and non-oil sources. Although Dávila-Santiago et al. (2022) recently reported an ML workflow utilizing support vector classification for selecting the top n diagnostic chemical features using LC-HRMS-based non-target chemical data, there is still a need to develop more complementary robust ML workflow to trace pollution sources using large dataset, especially those obtained from GC-HRMS-based SNTS.

Therefore, in the present study, we aimed to investigate the distribution characteristics of pollutants in groundwater samples from historically contaminated landfill, agricultural, industrial, and oil sites in South Korea using suspect and non-target screening method by GC-QTOF/MS. In addition, a novel and robust systematic ML classification model workflow was developed to pinpoint diagnostic chemical fingerprints for predicting pollution sources in the various groundwater sampling regions. The developed ML model workflow in this study will be useful for studies attempting to extract diagnostic information from chemical data, chemical forensic investigations, and water quality monitoring campaigns.

2. Materials and methods

2.1. Chemicals and materials

To construct an in-house personal compounds database and library (PCDL), 418 organic compounds were purchased (See Table S1). These included alkylated polycyclic aromatic hydrocarbons (PAHs) and petroleum-related compounds ($n = 31$), benzotriazole ultraviolet stabilizers ($n = 8$), nitro-PAHs ($n = 11$), novel brominated flame retardants

($n = 7$), n -Alkanes ($n = 31$), organophosphate flame retardants ($n = 10$), Oxy-PAHs ($n = 9$), pesticides ($n = 114$), phenols ($n = 12$), polychlorinated biphenyls ($n = 62$), polychlorinated dibenzo- p -dioxins/furans ($n = 17$), polychlorinated naphthalenes ($n = 18$), polybrominated diphenyl ethers ($n = 30$), parent PAHs ($n = 21$), plasticizers ($n = 27$), synthetic musk compounds ($n = 3$), and volatile methyl siloxanes ($n = 7$). The pure standard mixtures of the above compound groups were bought from a number of companies, including AccuStandard® (New Haven, CT, USA), Dr. EhrenstorferTM GmbH (Augsburg, Germany), Wellington Laboratories (Guelph, ON, Canada), Tokyo Chemical Industry (Tokyo, Japan), and Toronto Research Chemicals (Toronto, ON, Canada). The following materials were also purchased: HPLC grade water, pesticide grade methanol, ethyl acetate, and dichloromethane from Fisher Chemical (Waltham, MA, USA) and Honeywell Burdick & Jackson (Muskegon, MI, USA); Glass fiber filters (GF/Fs; 47 mm) from Whatman (Maidstone, UK); Oasis® HLB cartridges (200 mg, 6 cc) from Waters (Milford, MA, USA).

2.2. Sampling campaign

A total of 78 groundwater samples were collected from four selected regions with unique contamination histories, namely LF ($n = 22$), OC ($n = 29$), AGR ($n = 15$), and IND ($n = 12$), in the republic of Korea between June 2021 and August 2022. The OC region was near a former camping site for U.S. troops where continuous oil spillage was reported to occur. The LF region was near a residential neighborhood and subway station formerly used as a waste landfill site in 1970. The AGR region is known for agricultural activities, especially livestock farming, and swine farms around the sampling points was reported to have illegally dumped about 8500 tons of livestock manure wastewater in a lava cave over a period of several years; an act which was revealed in 2017. The IND region constitutes an industrial complex with various industries, such as electronics, (semi-conductor), textile, pharmaceutical, batteries, chemicals, cement, etc. The groundwater samples were collected in 1 L amber glass bottles, immediately shipped in dry ice to the laboratory, and stored in the refrigerator at 4 °C until subsequent pretreatment and instrument analysis which was basically within a few days.

2.3. Sample pretreatment and instrument analysis

The obtained groundwater samples were pretreated according to a validated protocol detailed in a recently published study by our research group (Sim et al., 2022). Briefly, each GF/F filtered groundwater sample (500 mL) contained in a pre-cleaned glass bottle, was extracted by the solid-phase extraction (SPE) method using 6cc Oasis® HLB cartridges (200 mg) pre-conditioned as follows: 10 mL Methylene chloride : Ethyl acetate (1/1 v/v, pesticide grade) → 5 mL Methanol (pesticide grade) → 10 mL water (HPLC grade). The loaded cartridges were then dried under vacuum and eluted with 12 mL Methylene chloride : Ethyl acetate (1/1 v/v, pesticide grade). The suspended particles collected on the GF/F glass filters were also extracted using the ultrasonic extraction (USE) method (2 × 10 mL), and the pooled extracts from both SPE and USE methods were further concentrated to 200 μ L final volume (in hexane) under pure N₂ gas. HPLC water and unused GF/F filter were also pre-treated and used as procedural blanks. 1 μ L of each final sample extract was then injected into the gas chromatography quadrupole time-of-flight mass spectrometry (GC-QTOF/MS) instrument (Agilent; Santa Clara, CA, USA). Further information regarding the instrumental analysis conditions are detailed in the supporting information (Table S2).

2.4. SNTS approaches employed for compounds identification

The identification workflow of suspect and non-target screening in this study was based on the report of previous studies (Ekpe et al., 2022; Castro et al., 2019; Lee et al., 2019), and the overall scheme is presented in Fig. 1 below. For suspect screening analysis, an in-house compounds

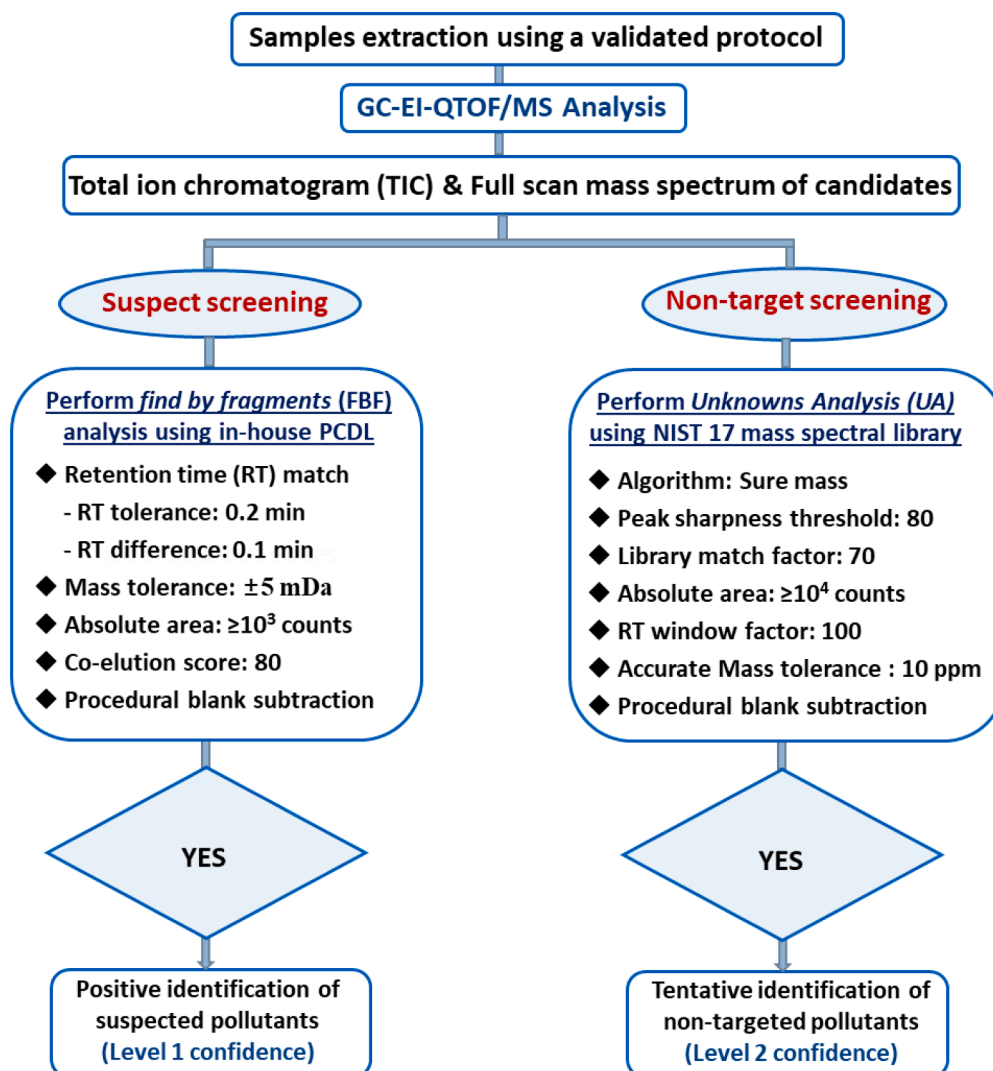


Fig. 1. Identification workflow of suspect and non-target screening.

database was constructed using the reference standards of 418 compounds (see Table S1 for the compounds list) and the chromatographic peak shape, retention times, and mass spectra of the compounds present in the samples extracts were matched with those stored in the database for the purpose of compounds identification based on the set of criteria outlined in Table S3. This was facilitated by the “Find Compounds by Fragments” workflow using Agilent’s *MassHunter Qualitative Analysis Workflows* software (ver. B.08.00). The detection of a reference ion and two or more qualifying ions in the sample extracts was regarded as a positive identification. Noticeable peak areas of compounds in the blank extracts were subtracted from the real samples extracts (Castro et al., 2019).

For non-target analysis, compounds not present in the in-house database but that may be present in the samples extracts were investigated using the NIST 17.L mass spectral database which contains nearly 1,000,000 spectra for 267,000+ compounds. This was facilitated by the “Sure Mass” algorithm using Agilent’s *MassHunter Unknowns Analysis* software (ver. B.09.00). The library spectra was compared against the sample data to identify and confirm the authenticity of the deconvoluted component hits, which were further manually inspected for good peak shape quality. The detailed set of criteria employed for compounds identification are provided in Table S4. Blank subtraction was carried out systematically (Lee et al., 2019).

2.5. Exploratory data analysis using statistical and machine learning tools

BoofTs package (<https://r-forge.r-project.org/projects/boofTs/>) was utilized for feature selection involving a combination of *rf_boruta* (i.e., random forests with Boruta feature selection algorithm), *PAMR* (Prediction analysis for microarrays, utilizes the nearest shrunken centroid classifier), and *GBM* (Gradient boosting machines, employs the gradient descent algorithm to minimize errors in sequential models), and these were implemented in R (version 4.2.2) (<http://www.R-project.org>) running in RStudio (version 2022.07.1 + 554) (<https://posit.co>). *MetImp* 1.2, an open source R-based web tool (accessible via <https://metabolomics.cc.hawaii.edu/software/MetImp/>) (Wei et al., 2018) was utilized for implementing *GSimp* (Gibbs sampler imputation) to resolve the issue of left-censored data. *MetaboAnalyst* 5.0, an open source R-based web tool (accessible via <https://www.metaboanalyst.ca/>) (Xia and Wishart, 2011) was utilized for multivariate analysis involving the partial least squares discriminant analysis (PLS-DA) as well as the Kruskal-Wallis test, whereby $p < 0.05$ was regarded as statistically significant. Correlation analysis was performed using the *ggcorrplot* R package. The top diagnostic chemical fingerprints were determined according to the PLS-DA variable important in prediction (VIP) scores, defined as the weighted, w_i^2 combination of the sum of squares of Y (i.e., outcome variable) explained by each latent variable, SSY_b , normalized to the cumulative sum of square, SSY_{cum} (Eq. (1)).

$$\sqrt{M \times \frac{\sum_{i=1}^K w_i^2 \times SSY_i}{SSY_{cum}}} \quad (1)$$

where, M is the total number of compounds, and K is the total number of latent variables.

3. Results and discussions

3.1. Identification of pollutants by suspect and non-target screening and their distribution characteristics

A total of 126 compounds were identified and confirmed in groundwater samples from four understudied sampling regions using the in-house personal compounds database and library (PCDL) suspect screening approach, while a total of 126 compounds were identified by the NIST 17.L mass spectral library non-target screening approach, summing up to 252 unique compounds in total. The identification confidence of the PCDL detected compounds was level 1 (i.e., retention time and spectral library match) while that of NIST 17.L detected compounds was level 2 (i.e., spectral library match), according to the criteria suggested by Schymanski et al. (2014). The identified compound groups ($n = 18$ in total) from both screening methods across the four understudied sampling regions consist of parent polycyclic aromatic hydrocarbons (PAHs) ($n = 21$), alkylated PAHs ($n = 26$), n -Alkanes ($n = 29$), heterocyclic PAHs ($n = 10$), hopanes ($n = 2$), diamondoids ($n = 2$), sesquiterpenes ($n = 5$), synthetic musks ($n = 2$), phenols ($n = 17$),

plasticizers ($n = 34$), benzotriazole ultraviolet stabilizers (BUVSs) ($n = 4$), volatile methyl siloxanes (VMSs) ($n = 15$), organophosphate flame retardants (OPFRs) ($n = 13$), pesticides ($n = 13$), pharmaceuticals ($n = 11$), personal care products (PCPs) ($n = 13$), metabolites ($n = 10$), and others ($n = 25$). Amongst these chemicals, 179 pollutants were identified in the oil-contaminated (OC) region at a detection frequency of 3–100 %, 154 pollutants were identified in the landfill (LF) region at a detection frequency of 5–100 %, 92 pollutants were identified in the agricultural (AGR) region at a detection frequency of 7–100 %, and 89 pollutants were identified in the industrial (IND) region at a detection frequency of 8–100 %. The full list of the identified individual compounds, including their abbreviations, molecular formula, retention times, m/z , CAS No., identification confidence level, and detection frequencies are shown in the supporting information (Table S5).

The distribution characteristics calculated by dividing the average peak area of each compound group by the sum of the average peak areas of all groups, showed some noticeable variation depending on the understudied region, with the exception of a few chemical groups that were commonly detected at similar or variable proportions across all the sampling regions, especially plasticizers, OPFRs, PCPs, and n -Alkanes, probably due to their ubiquitous nature. The sum of the proportion of plasticizers, OPFRs, PCPs, and n -Alkanes accounted for 20.3 %, 40.9 %, 46.6 %, and 63.1 % of the total distribution in the IND, OC, AGR, and LF regions, respectively (Fig. 2). The relatively high distribution of plasticizers and OPFRs across the understudied sampling regions can be attributed to the fact that they can easily undergo leaching and multi-media environmental transport during the manufacturing, handling,

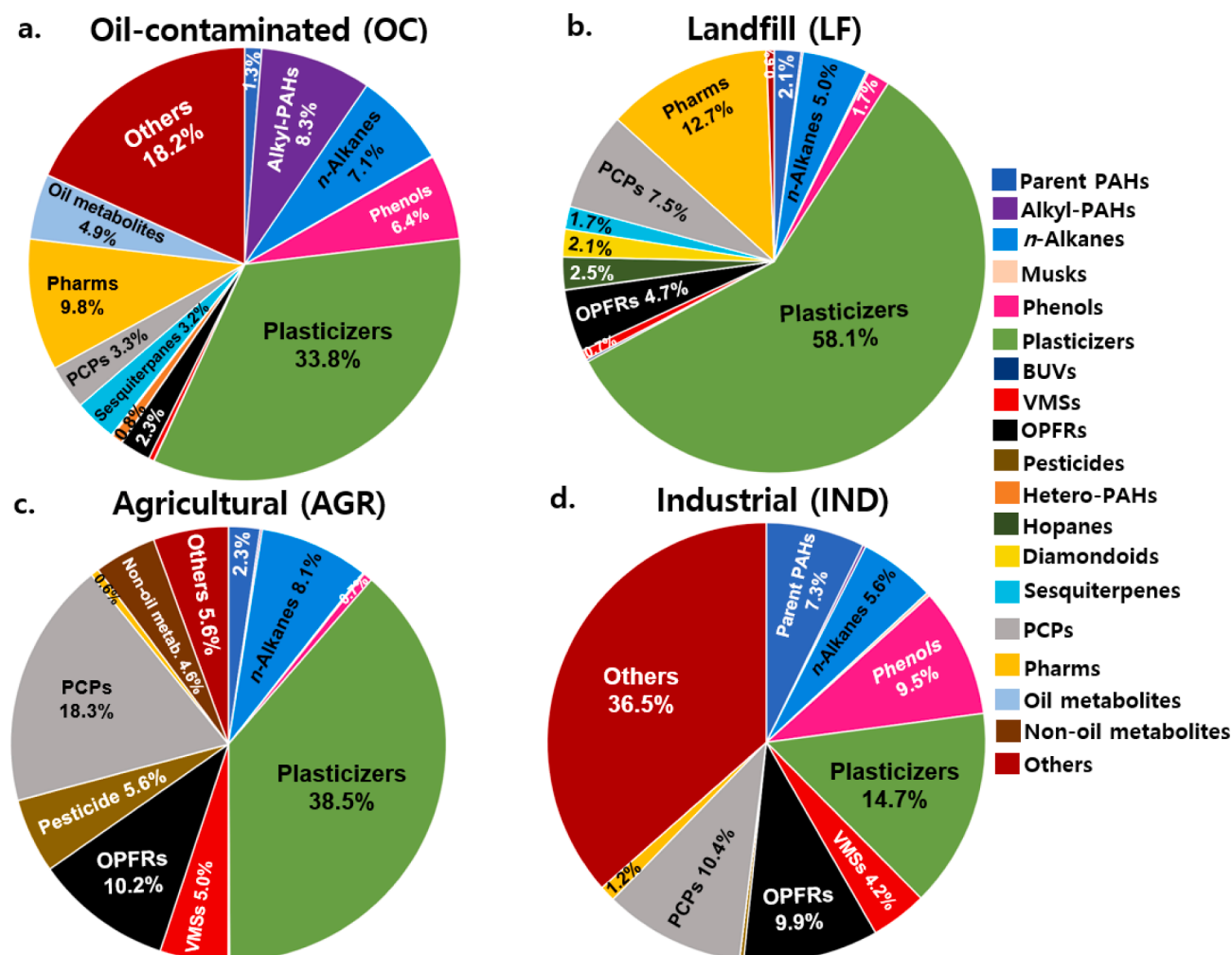


Fig. 2. Group-wise distribution of pollutants identified by GC-QTOF/MS suspect screening in the investigated sampling regions.

and disposal of their containing products, since they are often not chemically bonded to the polymer molecules of these materials (Liu et al., 2021; Sjödin et al., 2001). PCPs are often consumed in large quantities, and as such, may find their way directly or indirectly into the groundwater environment via various anthropogenic activities, such as landfill leachate, sewage discharge, and fertilizing (Sui et al., 2015). *n*-Alkanes have both biogenic or diagenetic and anthropogenic origins (e.g., incomplete combustion of fossil fuel and heavy oil emissions), justifying their widespread distribution in the environment (Liu et al., 2022).

In terms of the unique contamination profiles of each of the understudied regions, the OC region showed a unique distribution of alkyl-PAHs (8.3 %), oil metabolites (which comprised oxidation products, such as 2-indanol, 1-acenaphthenone, 1(2H)-Naphthalenone, 3,4-dihydro-, 1-adamantanol, 1-methyl-1-indanol, 1-methylindane-2-one, 7-methylindane-1-one, and o-Hydroxybiphenyl) (4.8 %), heterocyclic PAHs (0.8 %), and “others” (which comprised some petroleum-related chemicals, such as biphenyl, 1-methyl-Decahydronaphthalene, 4,9-dimethyl-Naphtho[2,3-b]thiophene, etc.) (18.2 %) in comparison to the other regions. In the AGR region, pesticides (5.6 %) and non-oil metabolites (which comprised 3,5-di-tert-Butyl-4-hydroxybenzaldehyde (a food additive metabolite) and phorate sulfone/phorate sulfoxide (insecticide metabolites)) (4.6 %) were the uniquely distributed pollutant groups. Due to the expected complexity of contamination characteristics in a typical landfill and industrial area, the LF and IND regions didn't show any obvious unique distribution in the compound groups, except for diamondoids (2.1 %) and hopanes (2.5 %) in the LF region and “others” (which comprised industrially relevant chemicals, such as ethylbenzene, butylated hydroxytoluene, styrene, tetrachloroethylene, succinimide, 2-ethylacrylate, etc.) (36.5 %) in the IND region. Furthermore, there were several noticeable overlap in the distribution profiles of the pollutant groups in terms of similarity, namely, pharmaceuticals and sesquiterpenes in the OC and LF regions, phenols in the OC and IND regions, VMSs in the AGR and IND regions (Fig. 2). Apparently, the distribution of petroleum related compounds, such as alkyl-PAHs, hetero-PAHs, C₁₁ and C₁₅ sesquiterpenes, C₂₉ and C₃₀ hopanes, diamondoids, etc. in the applicable regions suggests the possibility of oil spillage in those regions (Han et al., 2020; Kao et al., 2015; Yang et al., 2020; Yang et al., 2009). Nevertheless, it is evident from the above discussions that relying solely on the distribution characteristics of pollutant groups for predicting prevailing contamination sources may not be particularly useful. This is because environmental contamination profiles are complex, and as a result, it is essential to rely on statistically robust discriminatory methods for making such predictions.

3.2. Development and validation of ML based model to predict pollution sources

A typical mass spectrometry-based environmental or metabolomics dataset usually contains many zeros or “missing values” due to low detections of analytes in the sample matrix of interest. When these left-censored data are not handled properly prior to multivariate statistical analysis, bias in data interpretation is inevitable as they can affect the correlation between variables, thereby interfering with subsequent marker compounds discovery (Hrydziusko and Viant, 2012; Yang et al., 2015). In the present study, a combination of filter and robust imputation methods of data preprocessing was carried out to prevent possible deterioration in the performance of multivariate statistical analysis. Firstly, owing to the generally low detection frequency (DF) of compounds recorded in groundwater samples in this study (see Fig. S1 for the missingness map of the raw groundwater dataset), a 40 % group-wise DF cut-off was chosen as the filtering threshold beyond which the detection of analytes is considered to have left the window of chance (i.e., the predictive variables having a DF of at least 40 % in the samples of any one class, were proposed for non-removal from the dataset). This was done to decrease the risk of losing potentially important differential

chemical indicators which may have levels that are below the detection limit in any specific class(es), and the criteria was applied as an initial preprocessing step to exclude compounds that may be problematic to further data processing involving multivariate statistical techniques. As a result of the above data preprocessing step, the number of variables was dramatically reduced from 252 to 105 compounds (see Fig. S2 for the missingness map of the preprocessed groundwater dataset).

In the next step, GSimp was implemented to resolve the issue of non-detects (NDs) in the DF-filtered dataset. It enables the imputation of missing values while also taking into account the uncertainty in the imputed values. This approach helps to minimize bias in the downstream analyses, facilitating more accurate comparisons of data across various samples. Notably, GSimp have been reported to outperform other known methods in handling left-censored dataset with up to 60 % censoring rate at a relatively low normalized root mean square error (NRMSE) value (i.e., low imputation error) (Wei et al., 2018; Wang et al., 2023). It is particularly useful in HRMS-based metabolomics studies, where large amounts of data are often generated and missing values are common due to the limitations of the analytical methods used (Wei et al., 2018). The detailed step-wise framework of GSimp algorithm is described by Wei et al. (2018), and summarized in the supporting information (Text S1).

To identify indicators in each source, PLS-DA model was chosen in this study because of its intrinsic interpretability, resistance to multicollinearity, and ability to handle high-dimensional data (Gromski et al., 2015). Previous studies have reported that variable selection can help to improve the results of PLS-DA (Sorochan Armstrong et al., 2022; Kjeldahl and Bro, 2010); hence, in the present study, a “bootstrap selection” process was implemented using the *bootfs* package to obtain a robust ranking of the minimum number of compounds capable of providing the best classification performance across the sampling regions using the GSimp data. Three conceptually different machine learning algorithms (i.e., *rf*, *boruta*, *PAMR*, and *GBM*) (Bender, 2013), were combined to obtain a robust estimation of the importance rank of each compound capable of discriminating the four classes of samples in this study (i.e., LF, OC, AGR, and IND). The *bootfs* parameters are provided in the supporting information (Text S2), and the visual illustration of the *bootfs* workflow is also provided in the supporting information (Fig. S3). Prior to the bootstrap feature selection, a 5-fold cross-validation (repeated 10 times) of the selected *bootfs* methods was implemented to assess the individual performance of the classifiers, and the resulting median area under the curve (AUC) values were between 0.93 and 1.00 (see Fig. S4a). For a final selection of robust feature sets, the bootstrapping procedure was implemented at 100 iterations (repeated 20 times), and the relative selection frequency was used to rank the variables (see Fig. S4b). An 85 % threshold cut-off of the bootstrapped feature sets further reduced the number of compounds from 105 to 91, which were then used to develop the PLS-DA model with the aim of identifying the final set of source-relevant chemical indicators in the landfill, industrial, agricultural, and oil-contaminated areas.

The *bootfs*-processed groundwater data were first log₁₀-transformed and then auto-scaled to eliminate the dependence of the rank of compounds on the relative peak areas (van den Berg et al., 2006). To deal with possible overfitting, a robust resampling procedure known as the *k*-fold cross validation (where *k* = 5) was implemented, in which the samples were randomly divided into five approximately equal-sized folds; in each iteration, one fold was held out for testing or evaluating the PLS-DA model, while the remaining four fold was used for training the model. The PLS-DA model results showed a high “goodness of fit” (i.e., *R*²) of 0.932 and “goodness of prediction” (i.e., *Q*²) of 0.896 (Fig. 3b), indicating that the groundwater samples were well separated into different sampling class labels according to their pollution source characteristics (see Fig. 3a). Notably, the *Q*² and total explained variance (TEV) values of the non-*bootfs* applied PLS-DA were 0.876 and 46.4 %, respectively (see Fig. S5), which was lower than that of the *bootfs* applied PLS-DA (*Q*²=0.896, TEV=50.3 %), suggesting the importance of

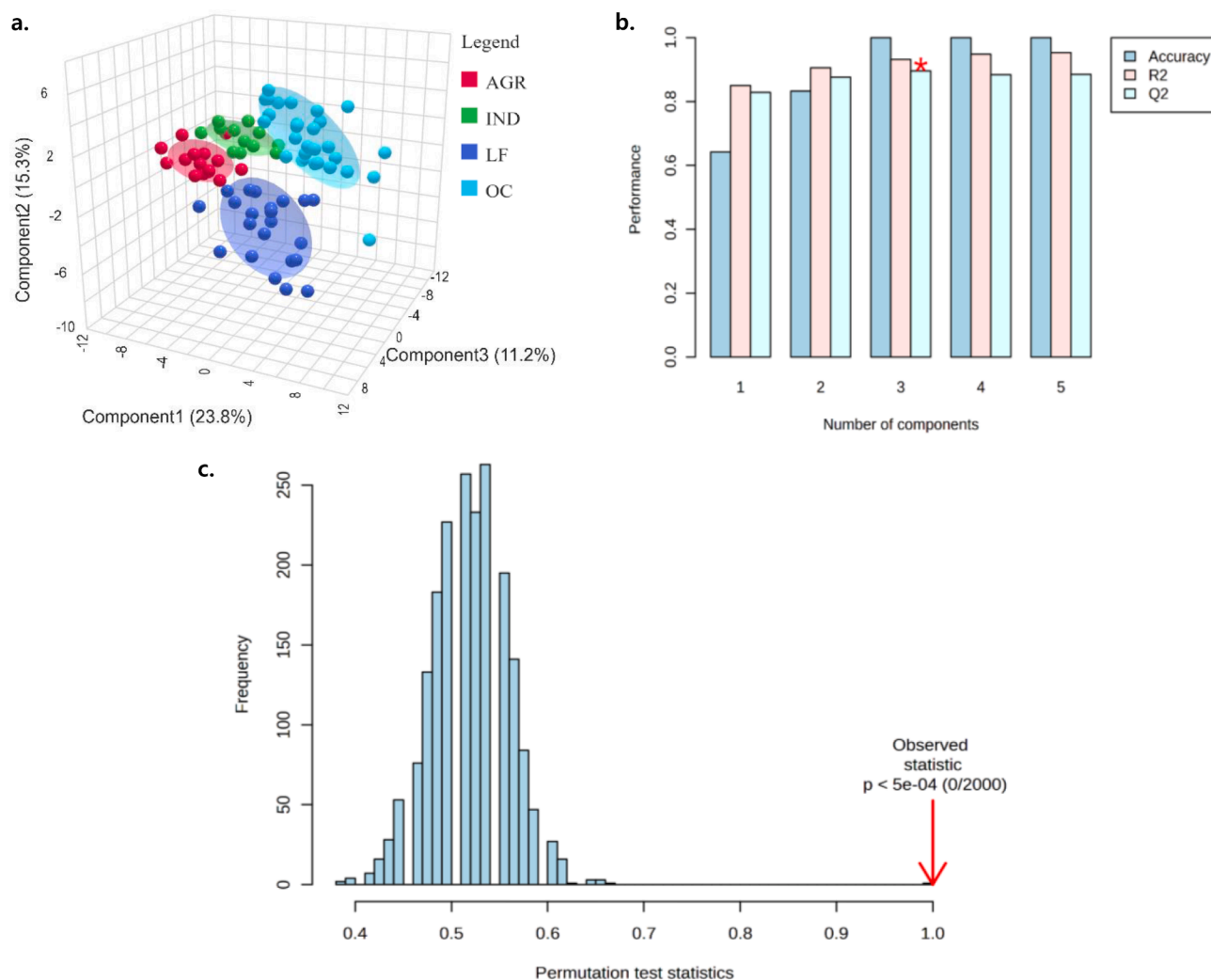


Fig. 3. (a) 3D scores plot of PLS-DA. The explained variances are shown in parenthesis; (b) Classification performance (based on a 5-fold cross validation) of PLS-DA using different number of components. The red asterisk indicates the best classifier; (c) Permutation test result of PLS-DA between the sample groups, performed using the optimal number of latent components at maximum Q^2 value.

feature selection step prior to PLS-DA, especially for a dataset with large number of redundant features contributing little or no unique information about the potential sources. The variables important in prediction (VIP) scores obtained from the proposed PLS-DA model were utilized for identifying the most important chemical indicators distinguishing each sample class based on a ≥ 0.8 threshold value (Song et al., 2022), and the suggested 51 chemical indicators are shown in Fig. 4. As expected, each of the top 51 diagnostic chemical fingerprints were statistically significantly higher in their respective predicted samples group (Kruskal-Wallis test, $p < 0.05$) (see Fig. S6).

Furthermore, although the obtained Q^2 value of 0.896 can be thought of as corresponding to a robust model with a high discriminating power, it is commonly argued that a high Q^2 value may be obtained purely by chance owing to the random nature of samples distribution in the training and testing sets of the PLS-DA model (Szymańska et al., 2012). Hence, to measure the statistical significance (i.e., p -value) of the model diagnostic statistics (i.e., Q^2 value), the class labels are often permuted and randomly assigned to different individual samples. A new classification model can then be calculated based on the ‘wrongly’ assigned class labels, with the assumption that no difference exists between the randomly formed groups. n -times repeated permutations of the procedure gives a non-significant null hypothesis (H_0) distribution of the model’s diagnostic statistics. The PLS-DA model’s statistical significance (at 95 % or 99 % confidence interval) is

determined by comparing the diagnostic statistics from the original (unpermuted) dataset to those from the permuted datasets (Szymańska et al., 2012). In the present study, an evaluation of the model’s validity based on a 2000 times permutation test showed that the PLS-DA model was statistically robust ($p < 0.0005$) (Fig. 3c), indicating the reliability of the 51 diagnostic chemical fingerprints identified by the developed PLS-DA model workflow.

3.3. Chemical indicators predicted from the developed ML based model

A summary of the chemical indicators predicted by the PLS-DA model and their known applications/uses are presented in Table 1. In the OC class, the predicted oil-related compounds suggests that the source of groundwater contamination in this region was mainly gasoline, diesel, and/or other light petroleum products. Their tendency to act as potential oil pollution indicators in the groundwater system may be attributed to their easy mobility in the soil-water system due to their high aqueous solubility, following an oil spillage accident (Duffy et al., 1980). MDHnap1 has been previously suggested as a potential petroleum biomarker for oil pollution source identification (Wang et al., 2015), although this study is the first to identify this compound in real environmental samples, namely groundwater. The prediction of alkyl-indane oxidation products (i.e., Ind1M2one, Ind7M1one) indicates the possibility of the weathering of spilled oil by bio- and/or

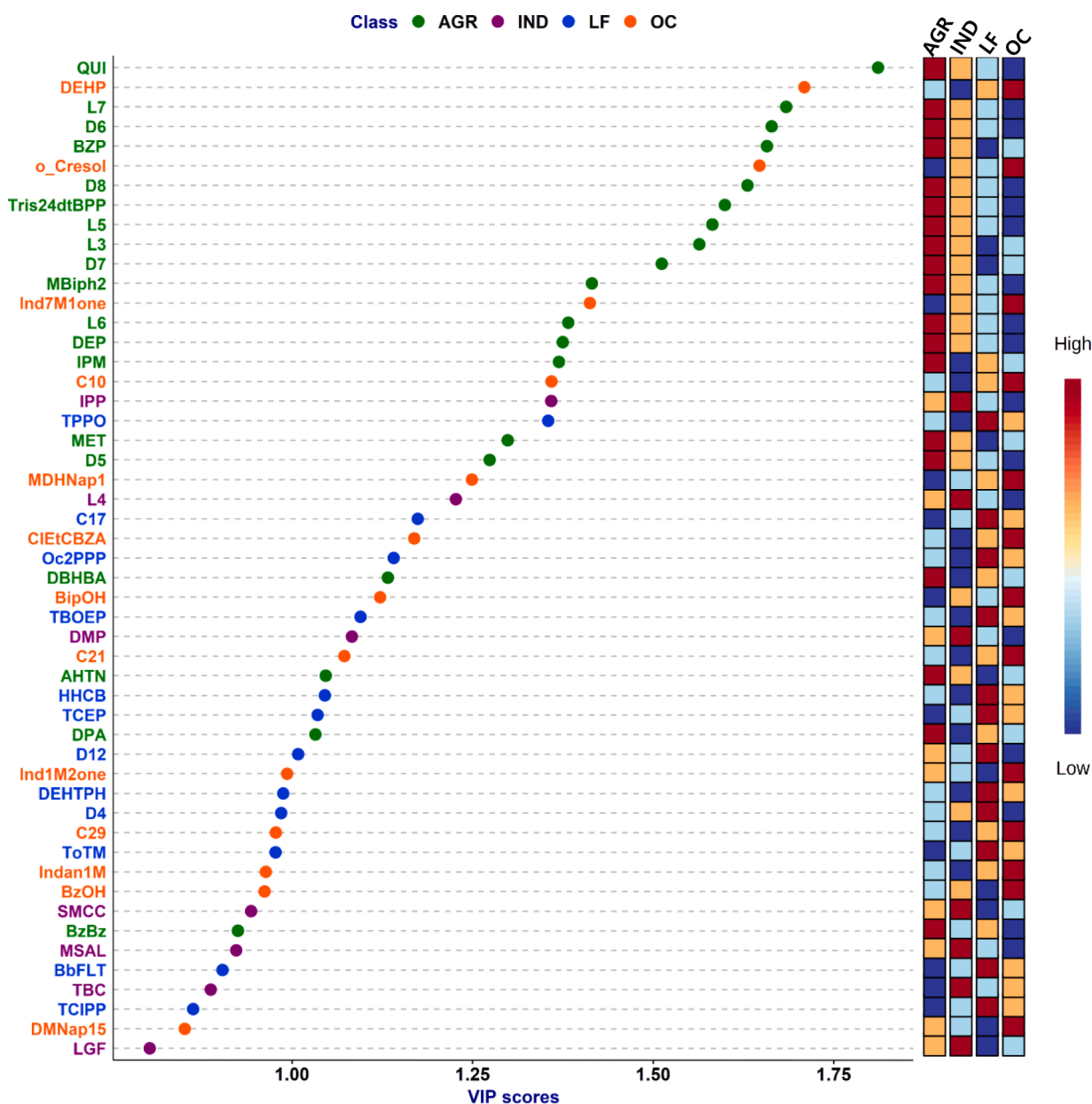


Fig. 4. Cleveland's dot plot shows the characteristic chemical indicators of groundwater dataset identified by PLS-DA VIP scores (≥ 0.8) for pollution source tracking (Higher VIP scores indicate greater influence on the model). The colored boxes on the right show the relative abundance of the corresponding compounds in each understudied sampling region (The different color patterns indicate a statistically significant difference at $p < 0.05$, Kruskal Wallis Test).

photo-degradation. Previous studies have reported that phenolic pollutants, including o-Cresol, are major components of shale oil wastewaters and oil shale leachates, which can potentially contaminate groundwater (Kahru et al., 2002; Yuting et al., 2021). Notably, the OC groundwater sampling points in the present study were near a former U. S. military base, and shale oil, a form of petroleum, is known to make up the largest proportion of the U.S. crude oil as about 7.23 million barrels per day (i.e., approx. 64 % of total U.S. crude oil) were produced from shale oil resources in 2021, according to the estimates of United States Energy Information Administration (EIA, 2022), suggesting that the source of o-Cresol in the oil-contaminated region of our study may likely originate from shale oil refined products, such as liquid petroleum gas (LPG), gasoline, and/or diesel fuel. In the case of *n*-Alkanes, their sources are relatively complex, but different compositions may indicate different sources. In the present study, the source of the predicted low molecular weight (LMW) *n*-Alkanes (i.e., C10 and C21) in the oil-contaminated region is believed to be from relatively fresh oil inputs

(and/or in the case of C21, old oil inputs due to its recalcitrance), especially light petroleum, such as gasoline or diesel which have abundant LMW *n*-Alkanes (Costa et al., 2022). Apparently, according to a 2018 report by Stars and Stripes, a US military's independent news source, the understudied oil polluted region was recently closed and has dozens of oil spill incidence history that is believed to have spanned a period of over 25 years, and seven of the accident involved a total spill volume of >3785 liters equivalent (Kim, 2023), thereby raising serious concerns over the long-term environmental impact on the overall groundwater quality owing to the possible persistence of the water-soluble components of spilled crude oil (Duffy et al., 1980). Besides the petroleum indicator compounds, the other contaminants (i.e., plasticizers and personal care products) also flagged as important compounds by the PLS-DA model, suggests that gasoline/diesel or other petroleum products can act as carrier solvents for these compounds.

Regarding the AGR class, 19 compounds were predicted by the PLS-DA model, typically consisting of pharmaceuticals, pesticides, a flame

Table 1

Summary of predicted chemical indicators in this study and their applications/uses.

Region	Chemical indicators	Class	HRMS ID conf. level	Applications/uses	Refs.
OC	1-Methyl-Decahydronaphthalene (MDHNap1)	Oil compound	Level 1	Previously suggested as a potential petroleum biomarker, but first identification in real environmental samples in the present study	(Wang et al., 2015)
	1,5-Dimethyl-Naphthalene (DMNap15)	Alkyl-PAHs	Level 1	Well-known representative PAHs in diesel fuel	(Kao et al., 2015)
	1-methylindan (Indan1M)	Oil compounds	Level 2	Known component of jet fuels	(Yu and Eser, 1998)
	1-Methylindan-2-one (Ind1M2one)	Oil metabolites	Level 2	Oxidation products of alkyl-indane	–
	7-Methylindan-1-one (Ind7M1one)		Level 2		
	2-Methylphenol (o-Cresol)	Phenol	Level 1	Major component of shale oil wastewater and refined products	(Yuting et al., 2021)
	o-Hydroxybiphenyl (BipOH)	Oil metabolite	Level 2	Desulfurization metabolite of Dibenzothiophene, a model S-containing compound commonly found in petroleum products.	(Nakayama et al., 2002)
	<i>n</i> -Decane (C10)	Low molecular weight (LMW) <i>n</i> -Alkanes	Level 1	May serve as indicators of relatively fresh oil inputs, especially light petroleum, such as gasoline or diesel. C21 is known to be recalcitrant, and hence may indicate old oil input	(Costa et al., 2022)
	<i>n</i> -Heneicosane (C21)				
	<i>n</i> -Nonacosane (C29)	High molecular weight (HMW) <i>n</i> -Alkane	Level 1	Known to mainly originate from plant litter	(Ren et al., 2020)
	2-((2-Chloroethoxy)carbonyl)benzoic acid (ClEtCBZA)	Plasticizers	Level 2	Utilized in plastics and cosmetic products	(Green et al., 2005)
	Bis(2-ethylhexyl) phthalate (DEHP)		Level 1		
	Benzyl alcohol (BzOH)	Personal care product (PCP)	Level 2	Utilized as a solvent and preservative in cosmetic formulations	(Nair et al., 2008)
AGR	Quinoline (QUI)	Pharmaceutical	Level 2	Nitrogen-containing compound known to possess strong pharmacological activity, including anti-fungal, anti-inflammatory, anti-malarial, anthelmintic, analgesic, anti-convulsant, anti-bacterial, and cardiotonic activity	(Marella et al., 2013)
	Heptasiloxane, hexadecamethyl- (L7)	VMSs	Level2	Typically used in body creams/lotions, cookware, cosmetics, hair creams, deodorants, and as silicone polymers production intermediate	(Lu et al., 2011)
	Cyclooctasiloxane, hexadecamethyl (D8)		Level 1		
	Cyclohexasiloxane, dodecamethyl- (D6)		Level 1		
	Pentasiloxane, dodecamethyl- (L5)		Level 2		
	Hexasiloxane, tetradecamethyl- (L6)		Level 2		
	Decamethylcyclopentasiloxane (D5)		Level 1		
	Cycloheptasiloxane, tetradecamethyl (D7)		Level 1		
	Trisiloxane, octamethyl- (L3)		Level 2		
	Diphenylamine (DPA)	Pesticides	Level 1	Priority pollutant widely used for preventing post-harvest deterioration of crops	(Drzyzga, 2003)
	Metolachlor (MET)		Level 1	Important herbicide known to commonly contaminate groundwater	(Liu and Xiong, 2009)
	3,5-di-tert-Butyl-4-hydroxybenzaldehyde (DBHBA)	Intermediate/ metabolite	Level 2	Used as an intermediate in the synthesis of pharmaceuticals and agrochemicals	(Nishizawa et al., 1983)
	Tris(2,4-di-tert-butylphenyl) phosphite (Tris24dtBPP)	OPFR	Level 2	Widely used antioxidant in plastics, such as food packaging materials	(Shi et al., 2020)
	Diethyl phthalate (DEP)	Plasticizers	Level 1	Findings application in plastics and a wide range of personal care products	(Weaver et al., 2020)
	Benzyl benzoate (BzBz)		Level 1		
	Isopropyl myristate (IPM)	Personal care product	Level 2	Typically used in cosmetics	(Nguyen et al., 2007)
	Benzophenone (BZP)		Level 2	Used in both cosmetics and food packaging materials	(Medina-Pérez et al., 2020)
	1,1'-Biphenyl, 2-methyl- (MBip2)	Other	Level 2	Not documented	–
	Tonalid (AHTN)	Synthetic musk	Level 1	Widely used in cosmetics and PCPs, such as shampoo, soaps, lotion, deodorants, detergents, air fresheners, and scented candles.	(Lu et al., 2011)
IND	Isopropyl palmitate (IPM)	PCP	Level 2	Emollient that is widely used in the manufacture of cosmetics and topical medicines	(Nguyen et al., 2007)
	Decamethyltetrasiloxane (L4)	VMS	Level 1	Ingredient in the production of silicon polymers and personal care products	(Lu et al., 2011)
	Butyl citrate (TBC)	PCP/Plasticizer	Level 2	Used as plasticizers in cosmetics and as intermediates in the production of resins	(Malnou, 2004; Young et al., 2018)
	Dimethyl phthalate (DM))		Level 1		
	Longifolene (LGF)	Sesquiterpene/ PCP	Level 2	A versatile sesquiterpene that finds wide-range applications in the flavour, chemical, perfumery, and pharmaceutical industries	–
LF	Methyl salicylate (MSAL)	Pharmaceutical	Level 2	Active ingredient in the manufacture of topical analgesic medicines commonly used for treating rheumatism, owing to its analgesic and anti-inflammatory properties	(Pauwels et al., 2012)
	Succinimide (SMCC)	Other	Level 2	Utilized as intermediates in various organic/pharmaceuticals syntheses and industrial silver-plating processes	–
	Cyclododecasiloxane, tetracosamethyl- (D12)	VMSs	Level 2	Mainly utilized in body creams/lotions, deodorants, and as intermediates in the manufacture of silicon polymers	(Lu et al., 2011)
	Cyclotetrasiloxane, octamethyl- (D4)				

(continued on next page)

Table 1 (continued)

Region	Chemical indicators	Class	HRMS ID conf. level	Applications/uses	Refs.
	Triphenylphosphine oxide (TPPO)	OPFRs	Level 2	Utilized as additive flame retardants and plasticizers in several household/industrial products	(Faiz et al., 2016)
	Tris(2-butoxyethyl) phosphate (TBOEP)		Level 1		
	Tri(2-chloroethyl) phosphate (TCEP)		Level 1		
	Tris(1-chloro-2-propyl) phosphate (TCIPP)		Level 1		
	Phthalic acid, octyl 2-propylpentyl ester (Oc2PPP)	Plasticizers	Level 2	Used as plasticizers in several consumer and industrial products, including toys, food contact materials, PVC/PVC co-polymers, flexible cable insulation materials, etc	–
	Bis(2-ethylhexyl) terephthalate (DEHTPH)		Level 2		
	Trioctyl trimellitate (ToTM)		Level 1	By-product of incomplete combustion of organic materials and gasoline/diesel exhaust	–
	Benzo[b]Fluoranthene (BbFLT)	PAH	Level 1		
	<i>n</i> -Heptadecane (C17)	LMW <i>n</i> -Alkane	Level 1	Plays a role as a plant metabolite and volatile oil component	PubChem
	Galaxolide (HHCB)	Synthetic musk	Level 1		
				Widely used in cosmetics and PCPs, such as shampoo, soaps, lotion, deodorants, detergents, air fresheners, and scented candles.	(Lu et al., 2011)

retardant, personal care products, and plasticizers, indicating that the source of groundwater contamination in this region is mainly from agricultural and human activities. One of the predicted compounds in this group (i.e., DBHBA) has been previously reported as one of the contaminants detected in German groundwater, with wet deposition being one of the suggested possible sources (Fries and Püttmann, 2004). Besides pesticides which are well-known groundwater pollutants in agricultural areas, the sources of PCPs (especially VMSSs) and plasticizers in this region may be attributable to livestock manure/wastewater, domestic wastewater, and agricultural plastic films used for greenhouse farming (Wang et al., 2021). In the case of the IND class, 7 compounds which are widely used as plasticizers, pharmaceuticals, and personal care products were predicted by the PLS-DA model, suggesting that the main source of groundwater contamination in this region was most likely from industrial and human activities. Finally, the sources of contamination in the groundwater from the LF area, being an historical landfill area, can be expected to be ubiquitous, including household, industrial, agricultural, construction, etc. Once in the landfill, pollutants contained in the landfill leachate can penetrate into the subsurface environment and contaminate groundwater resources via precipitation and/or surface runoff, thereby potentially affecting human life (Iravani and Ravari, 2020). The compounds ($n = 12$) predicted by the PLS-DA model in the LF region were comprised of personal care products, plasticizers, flame retardants, pesticides, pharmaceuticals, and combustion byproducts, indicating that the contamination source of pollutants in this region was mainly from anthropogenic activities.

3.4. Conclusions and perspectives

In this study, a comprehensive workflow integrating suspect and non-target screening by GC-QTOF/MS with machine learning methods was utilized for identifying organic micropollutants in the groundwater samples from diverse regions, namely oil, waste landfill, agricultural, and industrial. A total of 252 compounds were detected, including parent PAHs, alkylated PAHs, *n*-Alkanes, heterocyclic PAHs, hopanes, diamondoids, sesquiterpenes, synthetic musks, phenols, plasticizers, BUVs, VMSSs, OPFRs, pesticides, pharmaceuticals, personal care products, metabolites, and others. Amongst these identified pollutants, a bootstrapped feature selection method involving the *bootfs* algorithm, followed by a PLS-DA model was used to select each contamination source-relevant chemical indicators. A good predictive ability (Q^2) of 0.897 was obtained for the proposed model. Nevertheless, it should be noted that this current study only focused on mid- and non-polar compounds that could be readily resolved in gas chromatography. Consequently, future studies should take into account the analysis of polar compounds that may exhibit greater mobility in the environment using liquid chromatography when tracking pollution sources in groundwater.

Despite the limitation of using dataset obtained from only GC-HRMS, overall, the developed prediction workflow in this study was robust towards contamination source discrimination and may be adopted for environmental source chemical markers investigation in future studies, especially for cases involving relatively low compounds detection frequencies. The significance of the proposed workflow is in the fact that it can determine which features of the chemical space are most diagnostic for a specific contamination source class.

CRedit authorship contribution statement

Okon Dominic Ekpe: Investigation, Conceptualization, Visualization, Software, Data curation, Writing – original draft. **Gyojin Choo:** Validation, Writing – review & editing. **Jin-Kyu Kang:** Validation, Writing – review & editing. **Seong-Taek Yun:** Conceptualization, Validation, Funding acquisition. **Jeong-Eun Oh:** Supervision, Conceptualization, Validation, Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by a grant from the Korea Environmental Industry & Technology Institute (KEITI) through “Development of environmental forensics based on high resolution mass spectrometer to trace sources of subsurface contamination Project” funded by Korean Ministry of Environment (RS-2021-KE001426) and Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (NRF-2021R1A6A1A03039572, NRF-2021R1A2C2006517).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.watres.2024.121130](https://doi.org/10.1016/j.watres.2024.121130).

References

- Bender C. bootfs - Bootstrapped feature selection, 2013.
- Castro, G., Rodríguez, I., Ramil, M., Cela, R., 2019. Assessment of gas chromatography time-of-flight mass spectrometry for the screening of semi-volatile compounds in indoor dust. *Sci. Total Environ.* 688, 162–173. <https://doi.org/10.1016/j.scitotenv.2019.06.192>.
- Costa, C., Santos, A., Vega, M.A., 2022. Kinetics of Arab light crude oil degradation by *Pseudomonas* and *Bacillus* strains. *Water* 14, 3802. <https://doi.org/10.3390/w14233802>.
- Dávila-Santiago, E., Shi, C., Mahadwar, G., Medeghini, B., Insinga, L., Hutchinson, R., et al., 2022. Machine learning applications for chemical fingerprinting and environmental source tracking using non-target chemical data. *Environ. Sci. Technol.* 56, 4080–4090. <https://doi.org/10.1021/acs.est.1c06655>.
- Drzyzga, O., 2003. Diphenylamine and derivatives in the environment: a review. *Chemosphere* 53, 809–818. [https://doi.org/10.1016/S0045-6535\(03\)00613-1](https://doi.org/10.1016/S0045-6535(03)00613-1).
- Duffy, J.J., Peake, E., Mohtadi, M.F., 1980. Oil spills on land as potential sources of groundwater contamination. *Environ. Int.* 3, 107–120. [https://doi.org/10.1016/0160-4120\(80\)90045-8](https://doi.org/10.1016/0160-4120(80)90045-8).
- Ekpe, O.D., Choo, G., Choi, Y., Jeon, J., Oh, J.-E., 2022. Long-term degradation of toluene and phenol in soil: identification of transformation products and pathways via HRMS-based suspect and non-target screening. *J. Hazard Mater.* 430, 128429. <https://doi.org/10.1016/j.jhazmat.2022.128429>.
- Faiz, Y., Zhao, W., Feng, J., Sun, C., He, H., Zhu, J., 2016. Occurrence of triphenylphosphine oxide and other organophosphorus compounds in indoor air and settled dust of an institute building. *Build. Environ.* 106, 196–204. <https://doi.org/10.1016/j.buildenv.2016.06.022>.
- Fries, E., Püttmann, W., 2004. Monitoring of the antioxidant BHT and its metabolite BHT-CHO in German river water and ground water. *Sci. Total Environ.* 319, 269–282. [https://doi.org/10.1016/S0048-9697\(03\)00447-9](https://doi.org/10.1016/S0048-9697(03)00447-9).
- Gago-Ferrero, P., Schymanski, E.L., Hollender, J., Thomaidis, N.S., 2016. Chapter 13 - Nontarget analysis of environmental samples based on liquid chromatography coupled to high resolution mass spectrometry (LC-HRMS). In: Pérez S, Eichhorn P, Barceló D, editors *Compr. Anal. Chem.* 71, 381–403. <https://doi.org/10.1016/b.s.coc.2016.01.012>. Elsevier.
- Green, R., Hauser, R., Calafat, A.M., Weuve, J., Schettler, T., Ringer, S., et al., 2005. Use of di(2-ethylhexyl) phthalate-containing medical products and urinary levels of mono(2-ethylhexyl) phthalate in neonatal intensive care unit infants. *Environ. Health Perspect.* 113, 1222–1225. <https://doi.org/10.1289/ehp.7932>.
- Gromski, P.S., Muhamadali, H., Ellis, D.I., Xu, Y., Correa, E., Turner, M.L., et al., 2015. A tutorial review: metabolomics and partial least squares-discriminant analysis – a marriage of convenience or a shotgun wedding. *Anal. Chim. Acta* 879, 10–23. <https://doi.org/10.1016/j.aca.2015.02.012>.
- Guppy, L., Uyttendaele, P., Villholth, K.G., Smakhtin, V.U., 2018. *Groundwater and Sustainable Development Goals: Analysis of Interlinkages*. United Nations University Institute for Water, Environment and Health. Hamilton, Canada.
- Han, Q., Li, M., Liu, X., Jiang, W., Shi, S., Tang, Y., et al., 2020. Fractionation of alkylated carbazoles in petroleum during subsurface migration: evidence from molecular simulation and application in sandstone reservoirs. *J. Pet. Sci. Eng.* 191, 107308. <https://doi.org/10.1016/j.petrol.2020.107308>.
- Hrydziusko, O., Viant, M.R., 2012. Missing values in mass spectrometry based metabolomics: an undervalued step in the data processing pipeline. *Metabolomics* 8, 161–174. <https://doi.org/10.1007/s11306-011-0366-4>.
- Iravani, A., Ravari, S.O., 2020. Types of contamination in landfills and effects on the environment: a review study. *IOP Conf. Ser. Earth Environ. Sci.* 614. <https://doi.org/10.1088/1755-1315/614/1/012083>, 012083.
- Kahru, A., Maloverjan, A., Sillak, H., Pöllumaa, L., 2002. The toxicity and fate of phenolic pollutants in the contaminated soils associated with the oil-shale industry. *Environ. Sci. Pollut. Res. Int. (Spec No 1)*, 27–33. <https://doi.org/10.1007/BF02987422>.
- Kalogiouri, N.P., Aalizadeh, R., Dasenaki, M.E., Thomaidis, N.S., 2020. Application of high resolution mass spectrometric methods coupled with chemometric techniques in olive oil authenticity studies - a review. *Anal. Chim. Acta* 1134, 150–173. <https://doi.org/10.1016/j.aca.2020.07.029>.
- Kao, N.-H., Su, M.-C., Fan, J.-R., Chung, Y.-Y., 2015. Identification and quantification of biomarkers and polycyclic aromatic hydrocarbons (PAHs) in an aged mixed contaminated site: from source to soil. *Environ. Sci. Pollut. Res.* 22, 7529–7546. <https://doi.org/10.1007/s11356-015-4237-9>.
- Kayastha, V., Patel, J., Kathrani, N., Varjani, S., Bilal, M., Show, P.L., et al., 2022. New Insights in factors affecting ground water quality with focus on health risk assessment and remediation techniques. *Environ. Res.* 212, 113171. <https://doi.org/10.1016/j.envres.2022.113171>.
- Kibbey, T.C.G., Jabrzenski, R., O'Carroll, D.M., 2021. Source allocation of per- and polyfluoroalkyl substances (PFAS) with supervised machine learning: classification performance and the role of feature selection in an expanded dataset. *Chemosphere* 275, 130124. <https://doi.org/10.1016/j.chemosphere.2021.130124>.
- Kim G. US military leaves rich history, environmental concerns as Yongsan move gains momentum. *Stars Stripes* 2018. https://www.stripes.com/theaters/asia_pacific/us-military-leaves-rich-history-environmental-concerns-as-yongsan-move-gains-momentum-1.514046 (accessed February 14, 2023).
- Kjeldahl, K., Bro, R., 2010. Some common misunderstandings in chemometrics. *J. Chemom.* 24, 558–564. <https://doi.org/10.1002/cem.1346>.
- Kurwadkar, S., 2017. Groundwater pollution and vulnerability assessment. *Water Environ. Res.* 89, 1561–1577. <https://doi.org/10.2175/106143017X15023776270584>.
- Lee, S., Kim, K., Jeon, J., Moon, H.-B., 2019. Optimization of suspect and non-target analytical methods using GC/TOF for prioritization of emerging contaminants in the Arctic environment. *Ecotoxicol. Environ. Saf.* 181, 11–17. <https://doi.org/10.1016/j.ecoenv.2019.05.070>.
- Liu, B., Jiang, T., Li, Z., Ge, W., Wu, J., Song, N., et al., 2021. Phthalate esters in surface sediments from fishing ports in Circum-Bohai-Sea region, China. *Mar. Pollut. Bull.* 171, 112782. <https://doi.org/10.1016/j.marpolbul.2021.112782>.
- Liu, H., Xiong, M., 2009. Comparative toxicity of racemic metolachlor and S-metolachlor to *Chlorella pyrenoidosa*. *Aquat. Toxicol.* 93, 100–106. <https://doi.org/10.1016/j.aquatox.2009.04.006>.
- Liu, Y., He, Y., Liu, Y., Tao, S., Liu, W., 2022. Assessing spatiotemporal sources of biogenic and anthropogenic sedimentary organic matter from the mainstream Haihe River, China: using n-alkanes as indicators. *Sci. Total Environ.* 834, 155382. <https://doi.org/10.1016/j.scitotenv.2022.155382>.
- Lu, Y., Yuan, T., Wang, W., Kannan, K., 2011. Concentrations and assessment of exposure to siloxanes and synthetic musks in personal care products from China. *Environ. Pollut.* 159, 3522–3528. <https://doi.org/10.1016/j.envpol.2011.08.015>.
- Malnou A.M. Nail enamel composition containing a primary film forming polymer, a citrate ester, and an aldehyde condensation product. US20040022749A1, 2004.
- Marella, A., Tanwar, O.P., Saha, R., Ali, M.R., Srivastava, S., Akhter, M., et al., 2013. Quinoline: a versatile heterocyclic. *Saudi Pharm. J.* 21, 1–12. <https://doi.org/10.1016/j.jsps.2012.03.002>.
- Masiá, A., Campo, J., Blasco, C., Picó, Y., 2014. Ultra-high performance liquid chromatography–quadrupole time-of-flight mass spectrometry to identify contaminants in water: an insight on environmental forensics. *J. Chromatogr. A* 1345, 86–97. <https://doi.org/10.1016/j.chroma.2014.04.017>.
- Mazur, D.M., Detenckuk, E.A., Sosnova, A.A., Artaev, V.B., Lebedev, A.T., 2021. GC-HRMS with complementary ionization techniques for target and non-target screening for chemical exposure: expanding the insights of the air pollution markers in Moscow Snow. *Sci. Total Environ.* 761, 144506. <https://doi.org/10.1016/j.scitotenv.2020.144506>.
- Medina-Pérez, N.I., Arrizabalaga-Larrañaga, A., Seró, R., Moyano, E., 2020. Determination of benzophenone and related compounds in plastic packaged baby food by ultra-high-performance liquid chromatography coupled to tandem mass spectrometry. *Anal. Methods* 12, 358–367. <https://doi.org/10.1039/C9AY02134E>.
- Moschet, C., Lew, B.M., Hasenbein, S., Anumol, T., Young, T.M., 2017. LC- and GC-QTOF-MS as complementary tools for a comprehensive micropollutant analysis in aquatic systems. *Environ. Sci. Technol.* 51, 1553–1561. <https://doi.org/10.1021/acs.est.6b05352>.
- Nair, C.I., Jayachandran, K., Shashidhar, S., 2008. Biodegradation of phenol. *Afr. J. Biotechnol.* 7, 4951–4958.
- Nakayama, N., Matsubara, T., Ohshiro, T., Moroto, Y., Kawata, Y., Koizumi, K., et al., 2002. A novel enzyme, 2'-hydroxybiphenyl-2-sulfinate desulfinase (DszB), from a dibenzothiophene-desulfurizing bacterium *Rhodococcus erythropolis* KA2-5-1: gene overexpression and enzyme characterization. *Biochim. Biophys. Acta BBA - Proteins Proteomics* 1598, 122–130. [https://doi.org/10.1016/S0167-4838\(02\)00365-5](https://doi.org/10.1016/S0167-4838(02)00365-5).
- Nguyen, S.H., Dang, T.P., Maibach, H.I., 2007. Comedogenicity in rabbit: some cosmetic ingredients/vehicles. *Cutan. Ocul. Toxicol.* 26, 287–292. <https://doi.org/10.1080/15569520701555383>.
- Nishizawa K., Hamada K., Aratani T. Process for the production of 4-hydroxybenzaldehyde derivatives. EP0012939B1, 1983.
- Pauwels, J., D'Autry, W., Van den Bossche, L., Dewever, C., Forier, M., Vandenwaeyenberg, S., et al., 2012. Optimization and validation of liquid chromatography and headspace-gas chromatography based methods for the quantitative determination of capsaicinoids, salicylic acid, glycol monosacrylate, methyl salicylate, ethyl salicylate, camphor and l-menthol in a topical formulation. *J. Pharm. Biomed. Anal.* 60, 51–58. <https://doi.org/10.1016/j.jpba.2011.10.024>.
- Peter, K.T., Tian, Z., Wu, C., Lin, P., White, S., Du, B., et al., 2018. Using high-resolution mass spectrometry to identify organic contaminants linked to urban stormwater mortality syndrome in Coho Salmon. *Environ. Sci. Technol.* 52, 10317–10327. <https://doi.org/10.1021/acs.est.8b03287>.
- Ren, L., Wang, Y., Kawamura, K., Bikkina, S., Haghipour, N., Wacker, L., et al., 2020. Source forensics of n-alkanes and n-fatty acids in urban aerosols using compound specific radiocarbon/stable carbon isotopic composition. *Environ. Res. Lett.* 15, 074007. <https://doi.org/10.1088/1748-9326/ab8333>.
- Rivera-Pérez, A., Romero-González, R., Garrido Frenich, A., 2022. Fingerprinting based on gas chromatography-Orbitrap high-resolution mass spectrometry and chemometrics to reveal geographical origin, processing, and volatile markers for thyme authentication. *Food Chem.* 393, 133377. <https://doi.org/10.1016/j.foodchem.2022.133377>.
- Schymanski, E.L., Jeon, J., Gulde, R., Fenner, K., Ruff, M., Singer, H.P., et al., 2014. Identifying small molecules via high resolution mass spectrometry: communicating confidence. *Environ. Sci. Technol.* 48, 2097–2098. <https://doi.org/10.1021/es5002105>.
- Shi, J., Xu, C., Xiang, L., Chen, J., Cai, Z., 2020. Tris(2,4-di-tert-butylphenyl)phosphate: an unexpected abundant toxic pollutant found in PM2.5. *Environ. Sci. Technol.* 54, 10570–10576. <https://doi.org/10.1021/acs.est.0c03709>.
- Sibley, R.K.S.P., Hall, K., Keen, P., Mattu, G., Len, B.L., 2002. Sources, pathways, and relative risks of contaminants in surface water and groundwater: a perspective prepared for the Walkerton inquiry. *J. Toxicol. Environ. Health A* 65, 1–142. <https://doi.org/10.1080/152873902753338572>.
- Sim, W., Choi, S., Lee, H.-J., Kim, K., Park, K., Oh, J.-E., 2022. Evaluation of sample preparation methods for suspect and non-target screening in water, sediment, and biota samples using gas chromatography coupled to high-resolution mass spectrometry. *Sci. Total Environ.* 849, 157835. <https://doi.org/10.1016/j.scitotenv.2022.157835>.
- Sjödin, A., Carlsson, H., Thuresson, K., Sjölin, S., Bergman, Å., Östman, C., 2001. Flame retardants in indoor air at an electronics recycling plant and at other work

- environments. *Environ. Sci. Technol.* 35, 448–454. <https://doi.org/10.1021/es000077n>.
- Song, H., Yoon, S.-R., Dang, Y.-M., Yang, J.-S., Hwang, I.M., Ha, J.-H., 2022. Nondestructive classification of soft rot disease in napa cabbage using hyperspectral imaging analysis. *Sci. Rep.* 12, 14707. <https://doi.org/10.1038/s41598-022-19169-6>.
- Sorochan Armstrong, M.D., de la Mata, A.P., Harynuk, J.J., 2022. Review of variable selection methods for discriminant-type problems in chemometrics. *Front. Anal. Sci.* 2, 867938. <https://doi.org/10.3389/frans.2022.867938>.
- Sui, Q., Cao, X., Lu, S., Zhao, W., Qiu, Z., Yu, G., 2015. Occurrence, sources and fate of pharmaceuticals and personal care products in the groundwater: a review. *Emerg. Contam.* 1, 14–24. <https://doi.org/10.1016/j.emcon.2015.07.001>.
- Szymańska, E., Saccenti, E., Smilde, A.K., Westerhuis, J.A., 2012. Double-check: validation of diagnostic statistics for PLS-DA models in metabolomics studies. *Metabolomics* 8, 3–16. <https://doi.org/10.1007/s11306-011-0330-3>.
- Thomas, D.H., Delfino, J.J., 1991. A gas chromatographic/chemical indicator approach to assessing ground water contamination by petroleum products. *Groundw. Monit. Remediat.* 11, 90–100. <https://doi.org/10.1111/j.1745-6592.1991.tb00398.x>.
- U.S. EIA. Frequently Asked Questions (FAQs) - U.S. Energy Information Administration (EIA). US Energy Inf Adm EIA 2022. <https://www.eia.gov/tools/faqs/faq.php?id=847&t=6> (accessed April 11, 2022).
- van den Berg, R.A., Hoefsloot, H.C., Westerhuis, J.A., Smilde, A.K., van der Werf, M.J., 2006. Centering, scaling, and transformations: improving the biological information content of metabolomics data. *BMC Genomics* 7, 142. <https://doi.org/10.1186/1471-2164-7-142>.
- Wang, H., Zhang, S., Weng, N., Zhang, B., Zhu, G., Liu, L., 2015. Discovery and identification of a series of alkyl decalin isomers in petroleum geological samples. *Analyst* 140, 4694–4701. <https://doi.org/10.1039/C4AN00815D>.
- Wang, J., Gong, X., Hu, M., Zhao, L., 2023. Improved GSimp: a flexible missing value imputation method to support regulatory bioequivalence assessment. *Ann. Biomed. Eng.* 51, 163–173. <https://doi.org/10.1007/s10439-022-03070-4>.
- Wang, X., Yu, N., Yang, J., Jin, L., Guo, H., Shi, W., et al., 2020. Suspect and non-target screening of pesticides and pharmaceuticals transformation products in wastewater using QTOF-MS. *Environ. Int.* 137, 105599. <https://doi.org/10.1016/j.envint.2020.105599>.
- Wang, Y., Wang, F., Xiang, L., Gu, C., Redmile-Gordon, M., Sheng, H., et al., 2021. Risk assessment of agricultural plastic films based on release kinetics of phthalate acid esters. *Environ. Sci. Technol.* 55, 3676–3685. <https://doi.org/10.1021/acs.est.0c07008>.
- Wang, Z., Fingas, M., Landriault, M., Sigouin, L., Grenon, S., Zhang, D., 1999. Source identification of an unknown spilled oil from Quebec (1998) by unique biomarkers and diagnostic ratios of “source-specific marker” compounds. *Environ. Technol.* 20, 851–862. <https://doi.org/10.1080/09593330.1999.9618838>.
- Weaver, J.A., Beverly, B.E.J., Keshava, N., Mudipalli, A., Arzuaga, X., Cai, C., et al., 2020. Hazards of diethyl phthalate (DEP) exposure: a systematic review of animal toxicology studies. *Environ. Int.* 145, 105848. <https://doi.org/10.1016/j.envint.2020.105848>.
- Wei, R., Wang, J., Jia, E., Chen, T., Ni, Y., Jia, W., 2018. GSimp: a Gibbs sampler based left-censored missing value imputation approach for metabolomics studies. *PLOS Comput. Biol.* 14, e1005973. <https://doi.org/10.1371/journal.pcbi.1005973>.
- Xia, J., Wishart, D.S., 2011. Metabolomic data processing, analysis, and interpretation using metaboanalyst. *Curr. Protoc. Bioinforma.* 34. <https://doi.org/10.1002/0471250953.bi1410s34>, 14.10.1–14.10.48.
- Yang, C., Lambert, P., Nguyen, M., Yang, Z., Hollebone, B.P., Fieldhouse, B., et al., 2020. Application of gas chromatography-high resolution quadrupole time-of-flight mass spectrometry in fingerprinting analysis of polycyclic aromatic sulfur heterocycles. *J. Chromatogr. A* 1630, 461577. <https://doi.org/10.1016/j.chroma.2020.461577>.
- Yang, C., Wang, Z., Hollebone, B.P., Brown, C.E., Landriault, M., 2009. Characteristics of bicyclic sesquiterpanes in crude oils and petroleum products. *J. Chromatogr. A* 1216, 4475–4484. <https://doi.org/10.1016/j.chroma.2009.03.024>.
- Yang, J., Zhao, X., Lu, X., Lin, X., Xu, G., 2015. A data preprocessing strategy for metabolomics to reduce the mask effect in data analysis. *Front. Mol. Biosci.* 2, 4. <https://doi.org/10.3389/fmolb.2015.00004>.
- Young, A.S., Allen, J.G., Kim, U.-J., Sellar, S., Webster, T.F., Kannan, K., et al., 2018. Phthalate and organophosphate plasticizers in nail polish: evaluation of labels and ingredients. *Environ. Sci. Technol.* 52, 12841–12850. <https://doi.org/10.1021/acs.est.8b04495>.
- Yu, J., Eser, S., 1998. Thermal decomposition of jet fuel model compounds under near-critical and supercritical conditions. 2. Decalin and Tetralin. *Ind. Eng. Chem. Res.* 37, 4601–4608. <https://doi.org/10.1021/ie980302y>.
- Yuting, F., Changbo, L., Guozheng, Z., Hui, L., Shuo, W., Hongzhu, X., 2021. Study on pollution characteristics of shale refinery wastewater. *E3S Web Conf.* 233, 01146. <https://doi.org/10.1051/e3sconf/202123301146>.
- Zhong, S., Zhang, K., Bagheri, M., Burken, J.G., Gu, A., Li, B., et al., 2021. Machine learning: new ideas and tools in environmental science and engineering. *Environ. Sci. Technol.* 55, 12741–12754. <https://doi.org/10.1021/acs.est.1c01339>.